

Graphical representation of data and Information:

A chart is a **graphical representation** of **data**, in which "the **data** is represented by symbols, such as bars in a bar chart, lines in a line chart, or slices in a pie chart". ... A **data** chart is a type of diagram or graph, that organizes and represents a set of numerical or qualitative **data**.

General Rules for Graphic Presentation of Data and Information

There are certain [guidelines](#) for an attractive and effective graphic presentation of data and information. These are as follows:

1. **Suitable Title** – Ensure that you give a suitable title to the graph which clearly indicates the subject for which you are presenting it.
2. **Unit of Measurement** – Clearly state the unit of measurement below the title.
3. **Suitable Scale** – Choose a suitable scale so that you can represent the entire data in an accurate manner.
4. **Index** – Include a brief index which explains the different colors and shades, [lines](#) and designs that you have used in the graph. Also, include a scale of interpretation for better understanding.
5. **Data Sources** – Wherever possible, include the sources of information at the bottom of the graph.
6. **Keep it Simple** – You should construct a graph which even a layman (without any exposure in the areas of statistics or mathematics) can understand.
7. **Neat** – A graph is a visual aid for the presentation of data and information. Therefore, you must keep it neat and attractive. Choose the right size, right lettering, and appropriate lines, colors, dashes, etc.

Advantages of Graphic Presentation of Data and Information

- **Easy to understand** – Diagrammatic data presentation makes it easier for a common man to understand the data. Diagrams are usually attractive and impressive and many [newspapers](#) and magazines use them frequently to explain certain facts or [phenomena](#). Modern advertising campaigns also use diagrams.
- **Simplified Presentation** – You can represent large [volumes](#) of complex data in a simplified and intelligible form using diagrams.
- **Reveals hidden facts** – When you classify and tabulate data, some facts are not revealed. Diagrammatic data presentation helps in bringing out these facts and also [relations](#).

- **Saves time**- Diagrams present data quickly.
- **Quick to grasp** – Usually, when the data is represented using diagrams, people can grasp it quickly.
- **Easy to compare** – Diagrams make it easier to compare data.
- **Universally accepted** – Almost all fields of study like [Business](#), [economics](#), social institutions, [administration](#), etc. use diagrams. Therefore, they have universal acceptability.

Limitations of Graphic Presentation of Data and Information

- A diagram lacks complete accuracy of facts.
- It depicts only a few selected characteristics of the data.
- We cannot use a diagram support of a statement.
- A diagram is not a substitute for tables.
- Usually, laymen find it difficult to understand and interpret a diagram.
- Typically, a diagram shows the unreasonable tendency of the data and the actual values are not clear.

Types for Graphic Presentation of Data and Information

i)Bar diagram ii) Line diagram iii) Pie diagram iv) Scatter diagram v)Histogram
vi)Frequency polygon vii)Cumulative frequency polygon viii)Relative frequency polygon.

Bar Diagram Definition

The pictorial representations of a grouped data, in the form of vertical or horizontal rectangular bars, where the lengths of the bars are equivalent to the measure of data, are known as bar graphs or bar charts.

The bars drawn are of uniform width, and the variable quantity is represented on one of the axes. Also, the measure of the variable is depicted on the other axes. The heights or the lengths of the bars denote the value of the variable, and these graphs are also used to compare certain quantities. The frequency distribution tables can be easily represented using bar charts which simplify the calculations and understanding of data.

Types of Bar Diagram

The bar graphs can be vertical or horizontal. The primary feature of any bar graph is its length or height. If the length of the bar graph is more, then the values are greater of any given data.

Bar graphs normally show categorical and numeric variables arranged in class intervals. They consist of an axis and a series of labelled horizontal or vertical bars. The bars represent

frequencies of distinctive values of a variable or commonly the distinct values themselves. The number of values on the x-axis of a bar graph or the y-axis of a column graph is called the scale.

The types of bar charts are as follows:

1. Vertical bar chart
2. Horizontal bar chart

Vertical Bar Graphs

When the grouped data are represented vertically in a graph or chart with the help of bars, where the bars denote the measure of data, such graphs are called vertical bar graphs. The data is represented along the y-axis of the graph, and the height of the bars shows the values.

Horizontal Bar Graphs

When the grouped data are represented horizontally in a chart with the help of bars, then such graphs are called horizontal bar graphs, where the bars shows the measure of data. The data is depicted here along the x-axis of the graph, and the length of the bars denote the values.

Vertical Bar Graphs

The following data gives the information of the number of children involved in different activities.

Activities	Dance	Music	Art	Cricket	Football
No. of Children	30	40	25	20	53

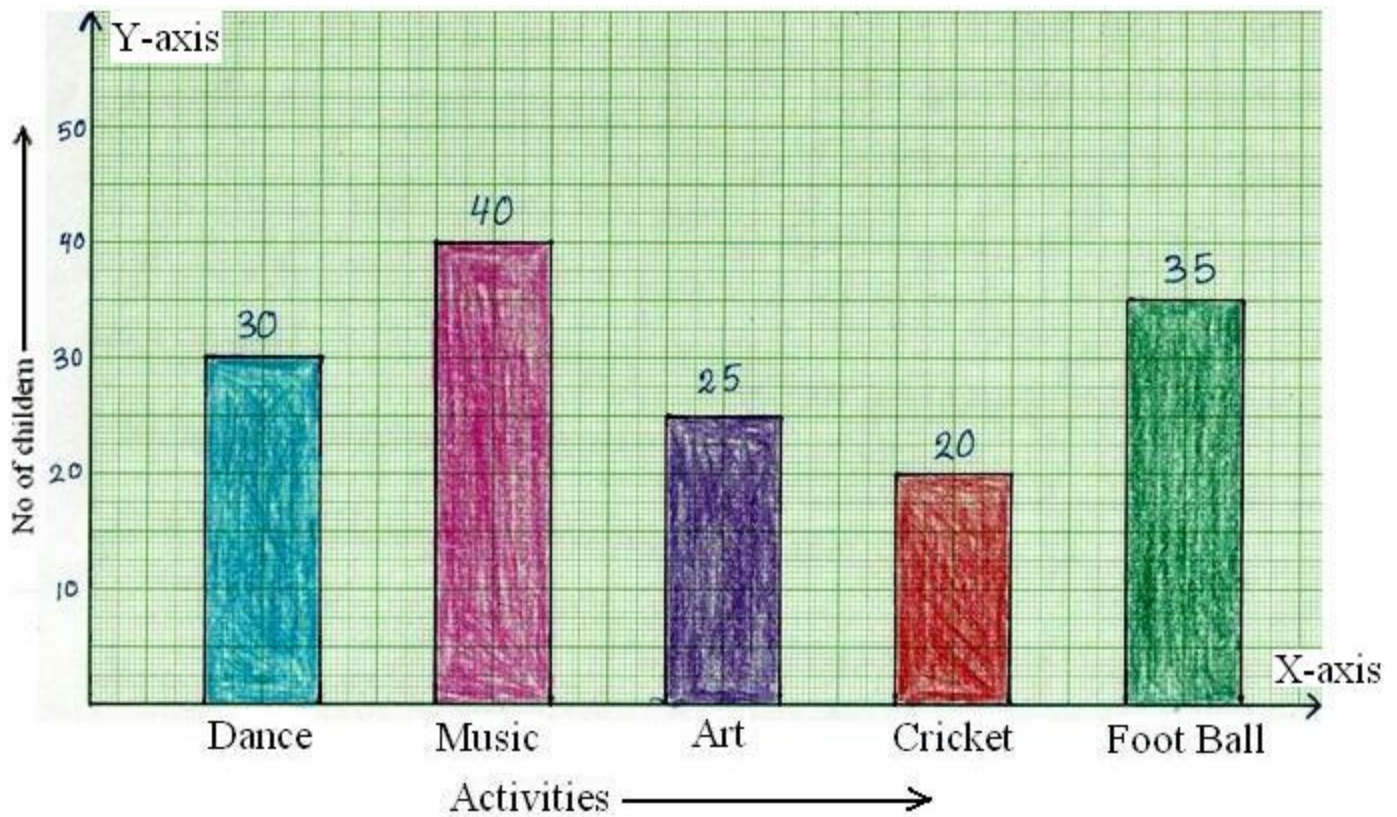
How to Construct a Bar Graph?

Steps in construction of bar graphs/column graph:

- On a graph, draw two lines perpendicular to each other, intersecting at 0.
- The horizontal line is x-axis and vertical line is y-axis.
- Along the horizontal axis, choose the uniform width of bars and uniform gap between the bars and write the names of the data items whose values are to be marked.
- Along the vertical axis, choose a suitable scale in order to determine the heights of the bars for the given values. (Frequency is taken along y-axis).

- Calculate the heights of the bars according to the scale chosen and draw the bars.

Bar graph gives the information of the number of children involved in different activities.

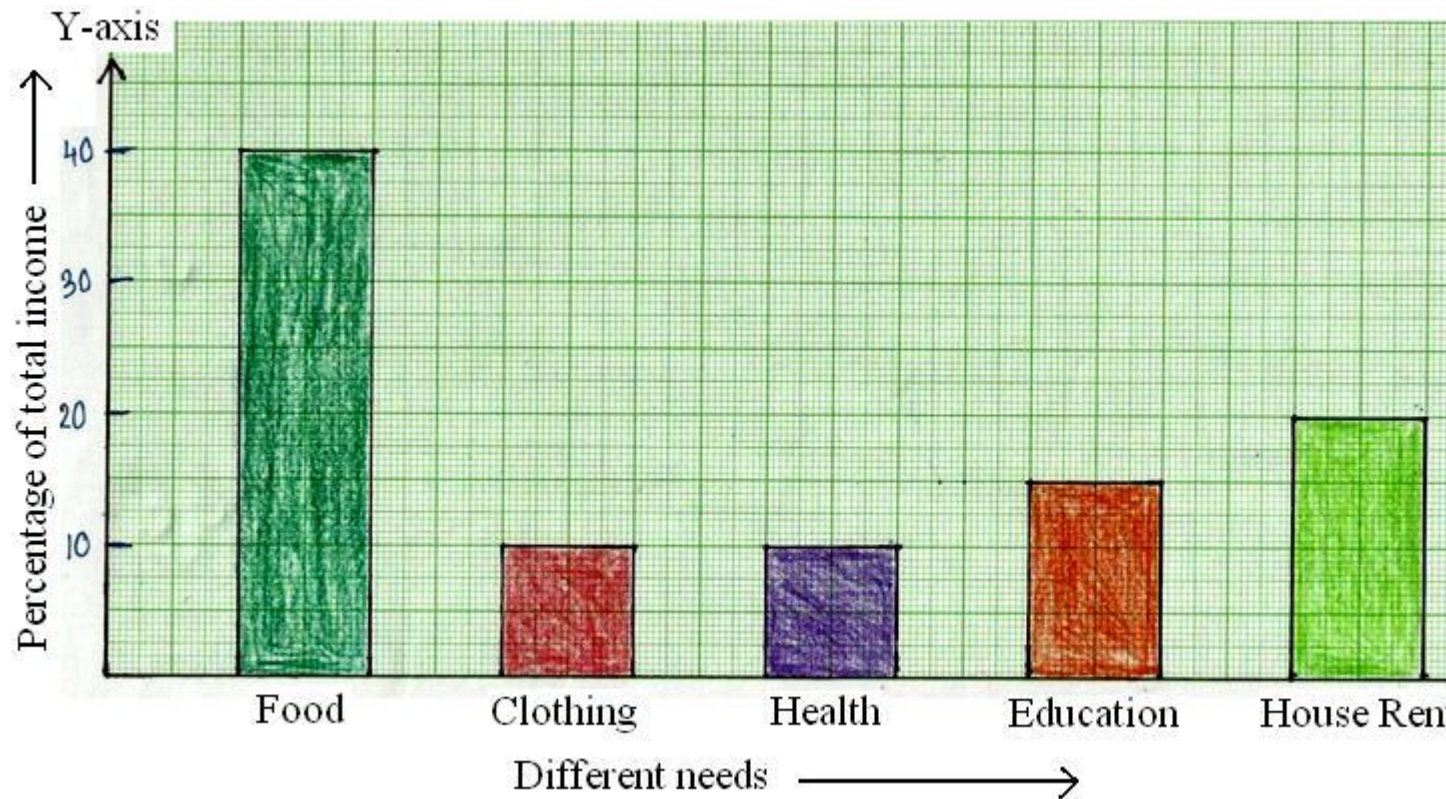


Solved examples on construction of bar graphs:

1. The percentage of total income spent under various heads by a family is given below.

Different Heads	Food	Clothing	Health	Education	House Rent	Miscellaneous
% Age of Total Number	40%	10%	10%	15%	20%	5%

Represent the above data in the form of bar graph.



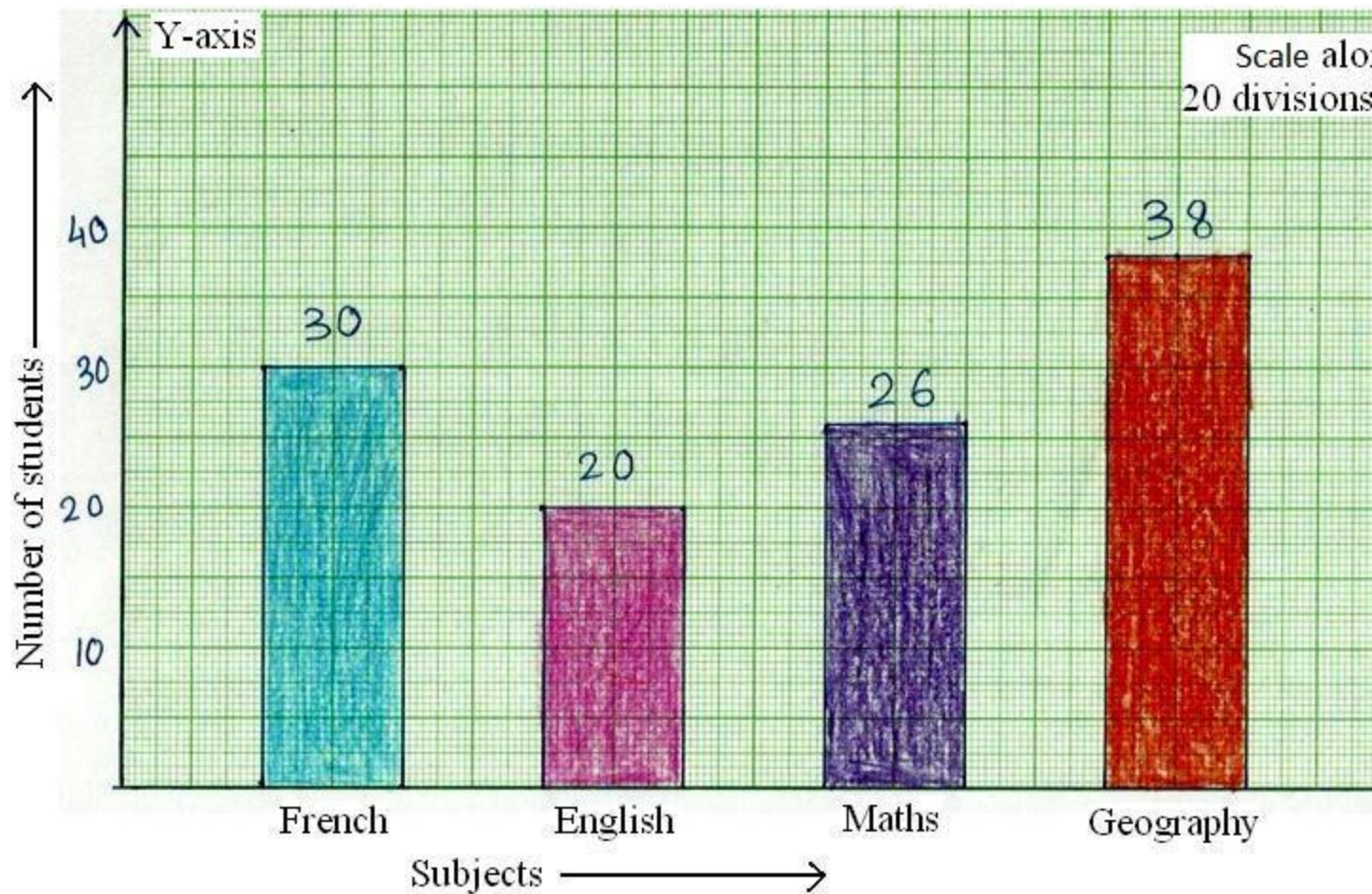
2. 150 students of class VI have popular school subjects as given below:

Subject	French	English	Maths	Geography	Science
Number of Students	30	20	26	38	34

Draw the column graph/bar graph representing the above data.

Solution:

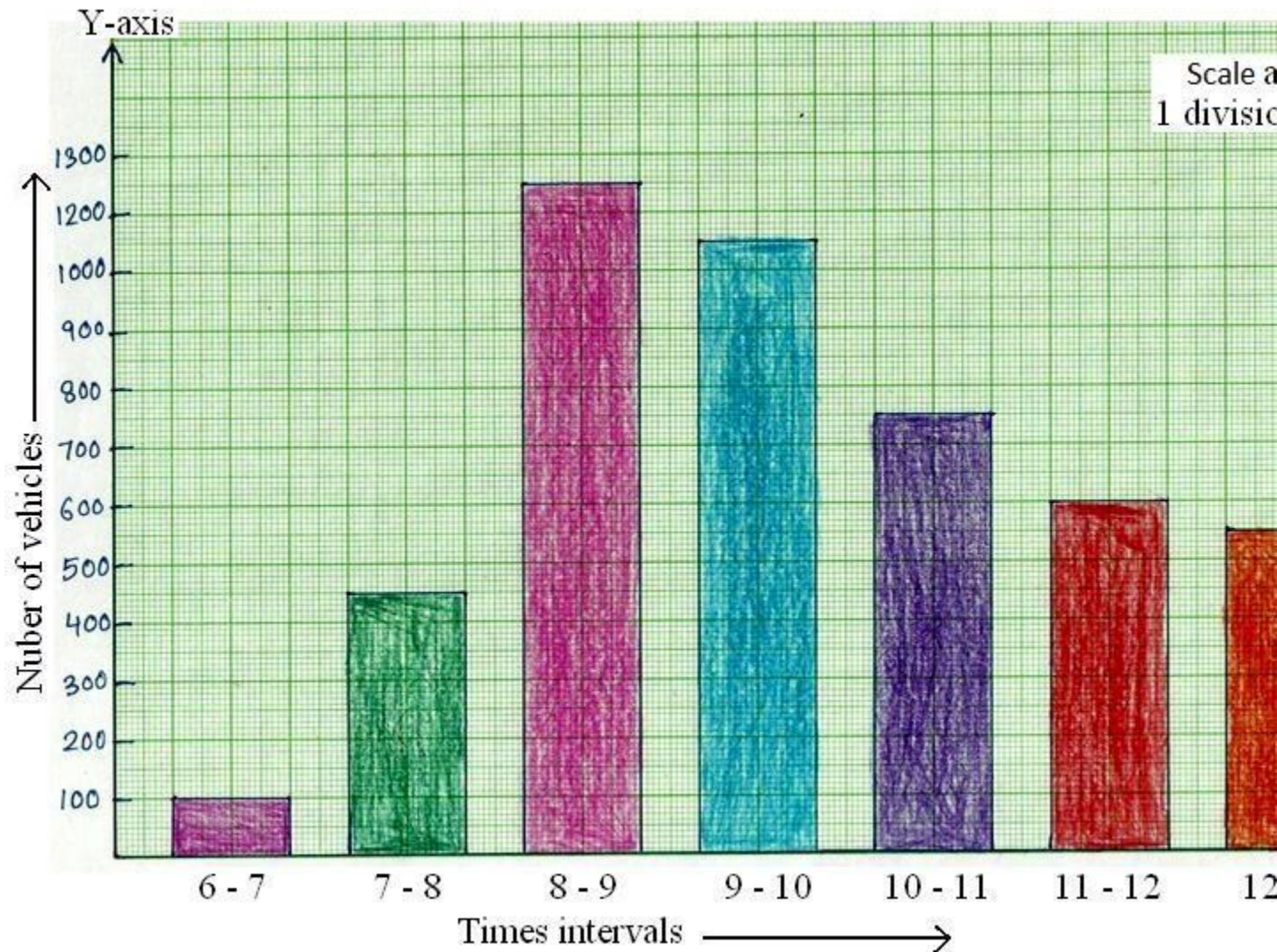
Take the subjects along **x-axis**, and the number of students along **y-axis**



Bar graph gives the information of favourite subjects of 150 students.

3. The vehicular traffic at a busy road crossing in a particular place was recorded on a particular day from 6am to 2 pm and the data was rounded off to the nearest tens.

Time in Hours	6 - 7	7 - 8	8 - 9	9 - 10	10 - 11	11 - 12	12 - 1	1 - 2
Number of Vehicles	100	450	1250	1050	750	600	550	200



Bar graph gives the information of number of vehicles passing through the crossing during different intervals of time.

Horizontal Bar Graph

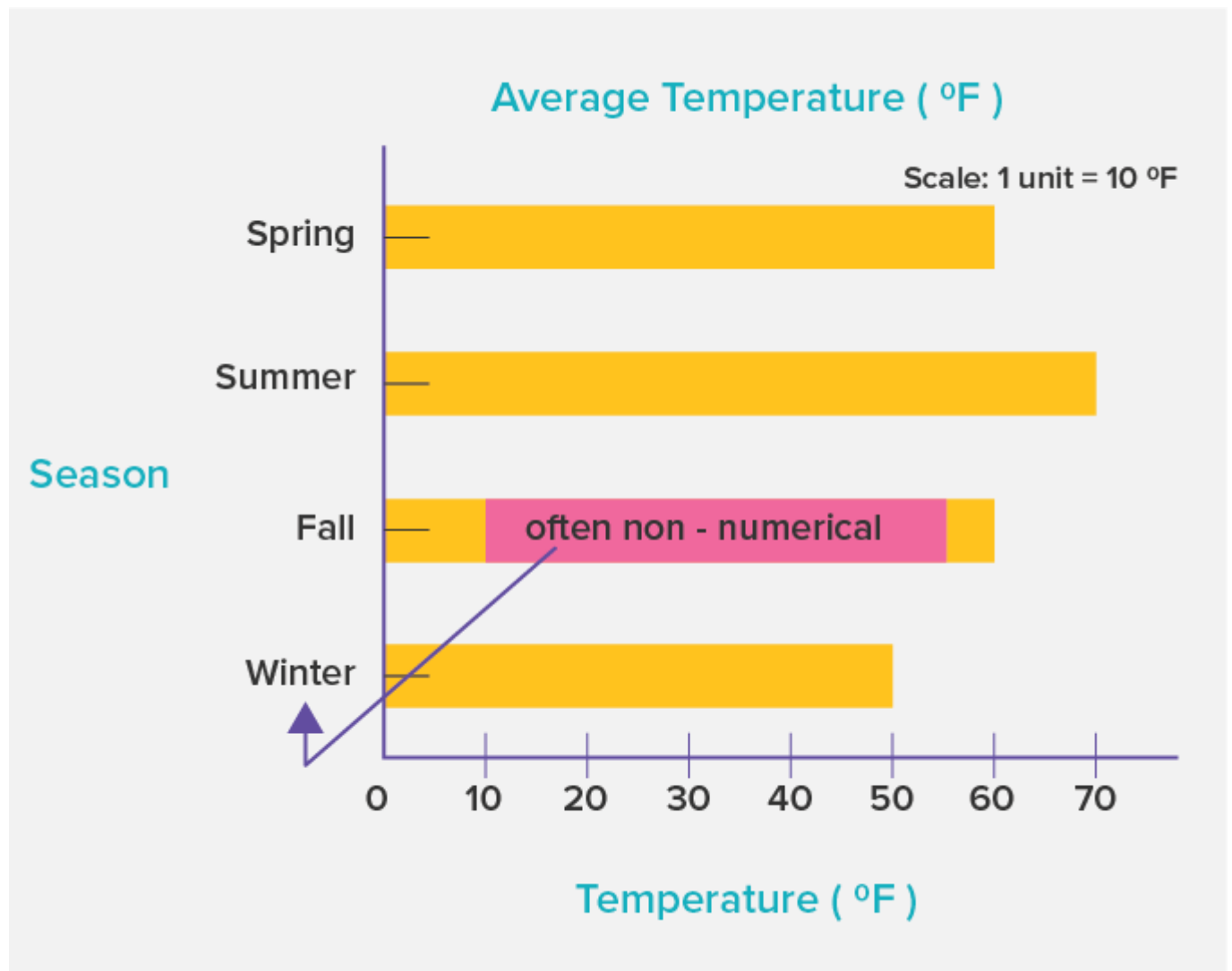
Bar Graph

A [bar graph](#) is a graph with rectangular bars with [lengths](#) and [heights](#) proportional to the values that they represent.

- On one axis of the [graph](#), it shows the data categories that are being compared.
- The other axis represents the values corresponding to each data category.

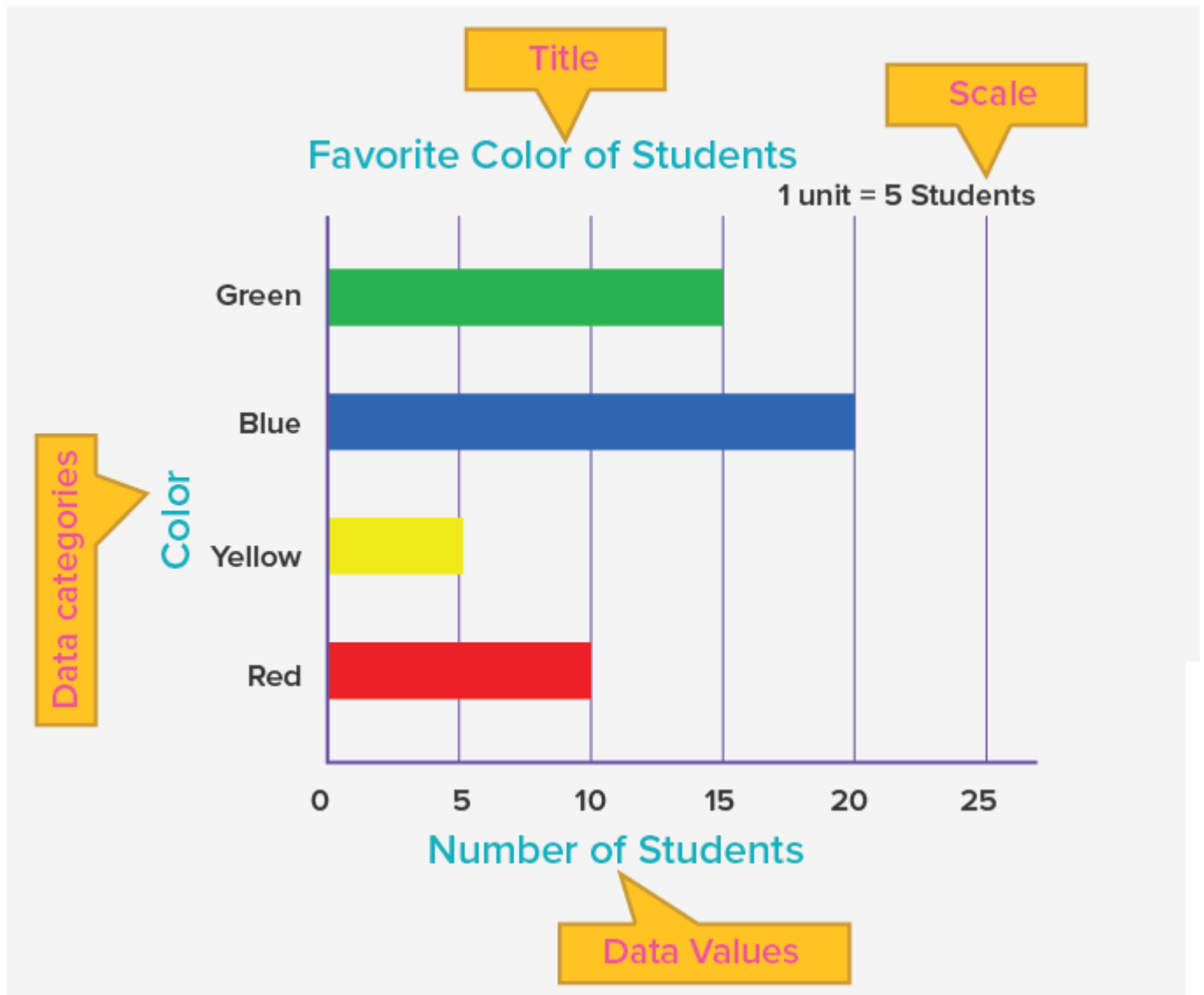
Horizontal bar graph

[Horizontal](#) bar graphs represent the data horizontally. It is a graph whose bars are drawn horizontally. The data categories are shown on the vertical axis and the data values are shown on the horizontal axis. The length of each bar is equal to the value corresponding the data category and all bars go across from left to right.



Reading a horizontal bar graph

- The title of the horizontal bar graph tells about the data being represented by the graph.
- The vertical axis represents the data categories. Here, the data categories are the colors.
- The horizontal axis represents the values corresponding to each data value. Here, the data values are the number of students who like a particular color represented on the vertical axis.
- The scale shows the value of 1 unit on the horizontal axis.



Line diagram

We may represent data using a line graph. A line graph is formed by joining the points given by the data with straight lines.

A **line graph** is usually used to show the **change of information over a period of time**. This means that the horizontal axis is usually a **time scale**, for example minutes, days, months or years.

Example:

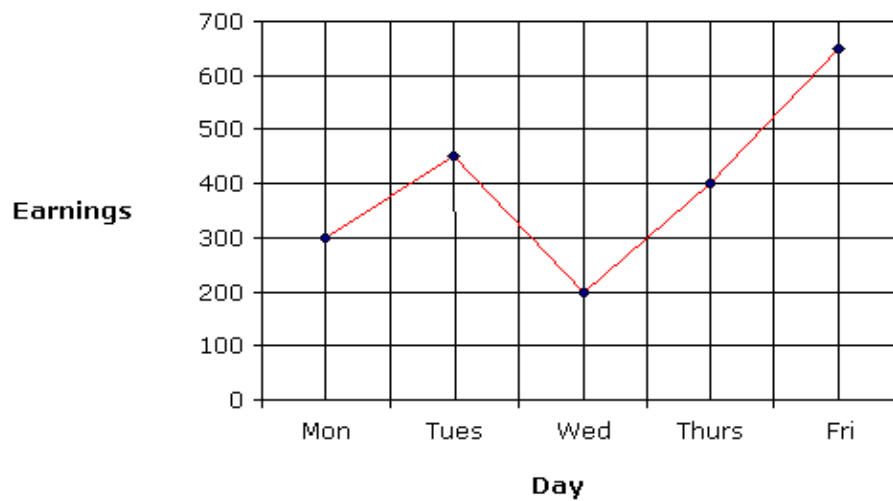
The table shows the daily earnings of a store for five days.

Day	Mon	Tues	Wed	Thurs	Fri
Earnings	300	450	200	400	650

- a) Construct a line graph for the table.
- b) On which days were the earnings above tk 400

Solution:

a)



- b) The earnings were above tk 400 on Tuesday and Friday.

Example:

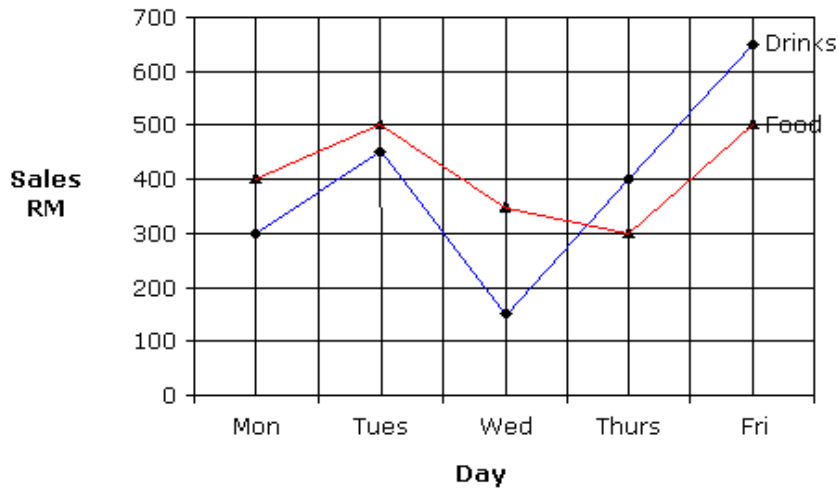
The table shows the daily sales in RM of different categories of items for five days.

Day	Mon	Tues	Wed	Thurs	Fri
Drinks	300	450	150	400	650
Food	400	500	350	300	500

- a) Construct a line graph for the table.
- b) On what days were the sales for drinks better than the sales for food?
- c) What is the total earnings for food and drinks on Wednesday?

Solution:

a)



b) Sales for drinks were better than sales for food on Thursday and Friday

c) Total earnings for food and drinks on Wednesday is

$$150 + 350 = \text{tk } 500$$

Pie diagram

A Pie Chart is a type of graph that displays data in a circular graph. The pieces of the graph are proportional to the fraction of the whole in each category. In other words, **each slice of the pie is relative to the size of that category** in the group as a whole.

Pie chart is best used when trying to work out the composition of something. If you have categorical data then using a pie chart would work really well as each slice can represent a different category. Another good use for a pie chart would be to compare areas of growth within a business such as turnover, profit and exposure.

A pie chart is a circular graph that is broken down into segments (i.e slices of pie). These segments represent each category's contribution to display parts of a whole. So if your data does not represent this then you should choose another chart. Also when there are similar sized categories, it can be difficult to interpret, so keep that in mind. Although they are very aesthetically pleasing and easy interpret and very powerful, so it is no wonder they are the most ubiquitous choice.

Now we will discuss about the construction of pie chart or pie graph. In brief let us recall about, what is a pie chart?

It is a circular graph which is used to represent data. In this :

- Various observations of the data are represented by the sectors of the circle.
- The total angle formed at the centre is 360° .
- The whole circle represents the sum of the values of all the components.
- The angle at the centre corresponding to the particular observation component is given by

$$\frac{\text{Value of the component}}{\text{Total value}} \times 360^\circ$$

Solved example on

construction of pie chart/pie graph:

1. The following table shows the numbers of hours spent by a child on different events on a working day.

Represent the adjoining information on a pie chart

Activity	No. of Hours
School	6
Sleep	8
Playing	2
Study	4
T. V.	1
Others	3

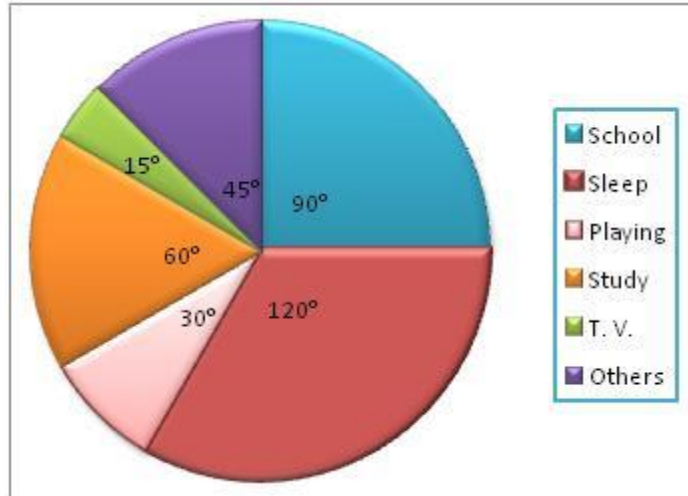
The central angles for various observations can be calculated as:

Activity	No. of Hours	Measure of central angle
School	6	$(\frac{6}{24} \times 360)^\circ = 90^\circ$
Sleep	8	$(\frac{8}{24} \times 360)^\circ = 120^\circ$
Playing	2	$(\frac{2}{24} \times 360)^\circ = 30^\circ$
Study	4	$(\frac{4}{24} \times 360)^\circ = 60^\circ$
T. V.	1	$(\frac{1}{24} \times 360)^\circ = 15^\circ$
Others	3	$(\frac{3}{24} \times 360)^\circ = 45^\circ$

Total

24

Now, we shall represent these angles within the circle as different sectors. Then we now make the pie chart:



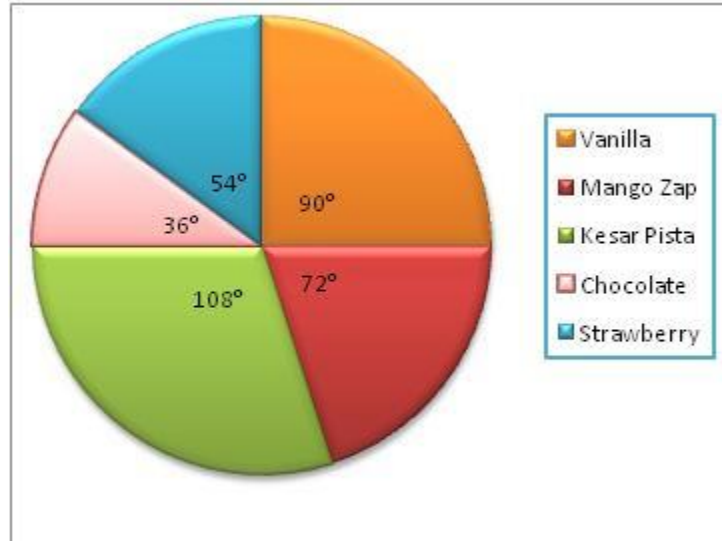
2. The favourite flavours of ice-cream for the children in a locality are given in percentage as follow. Draw the pie chart to represent the given information

Flavours	% of Students Prefer the Flavours
Vanilla	25 %
Strawberry	15 %
Chocolate	10 %
Kesar-Pista	30 %
Mango Zap	20 %

The central angles for various observations can be calculated as:

Flavours	% of Students Prefer the Flavours	Measure of Central Angles
Vanilla	25 %	$(\frac{25}{100} \times 360)^\circ = 90^\circ$
Strawberry	15 %	$(\frac{15}{100} \times 360)^\circ = 54^\circ$
Chocolate	10 %	$(\frac{10}{100} \times 360)^\circ = 36^\circ$
Kesar-Pista	30 %	$(\frac{30}{100} \times 360)^\circ = 108^\circ$
Mango Zap	20 %	$(\frac{20}{100} \times 360)^\circ = 72^\circ$

Now, we shall represent these angles within a circle to obtain the required pie graph.



Scatter Diagram

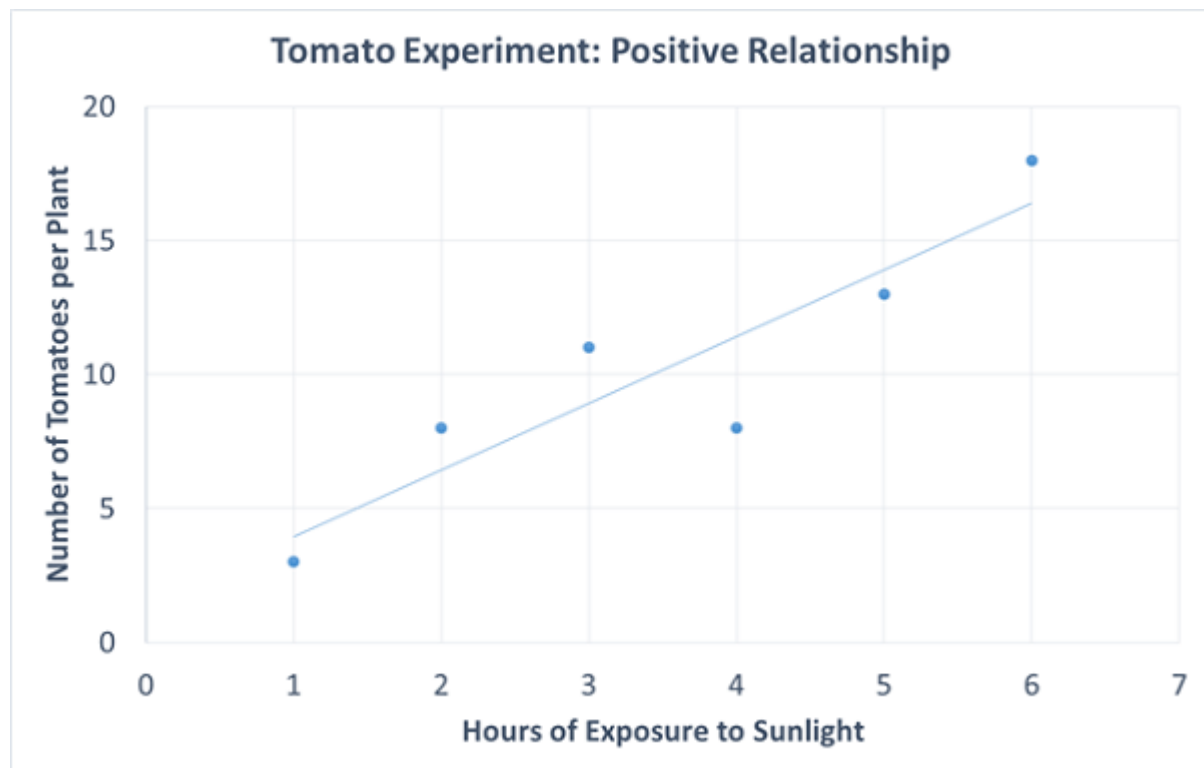
The **scatter diagram graphs** pairs of numerical data, with one variable on each axis, to look for a relationship between them. If the variables are correlated, the points will fall along a line or curve. The better the correlation, the tighter the points will hug the line. . We use Scatter Diagram when there is too much information whereas we can't afford enough time for bar chart - It is also better version of efficiency comparing to Bar Charts. - We use scatter diagrams like in stock market, Business Man - Company (IT people) - Government – Teachers.

Creating a Scatter Diagram

Number of Hours of Sunlight Exposure for Each Plant	Number of Tomatoes on Each Plant
"x"-axis	"y"-axis
1	3
2	8
3	11
4	8
5	13
6	18

Let's plot this data on (x, y) coordinates. We'll plot the hours of exposure to sunlight on the x-axis and the number of tomatoes for the same plant on the y-axis.

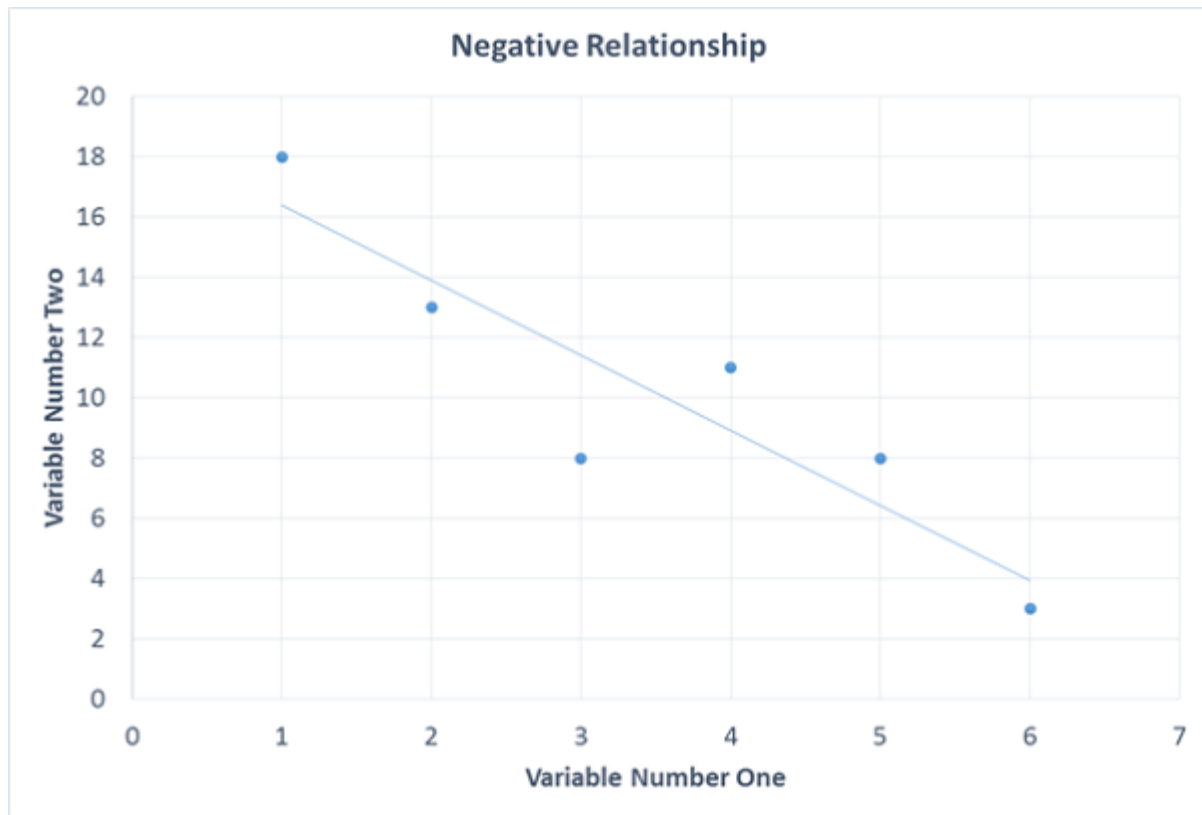
Now we've plotted each point on the graph appearing below.



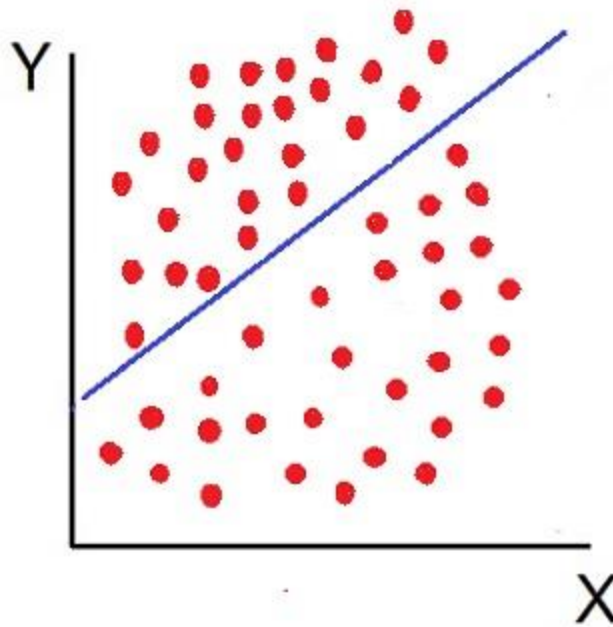
Notice that the number of tomatoes on a plant increases as the exposure to the sunlight increases. We can see this easily by drawing a trend line that best fits all your data points. Notice that the trend line is pointing in the northeast direction, which means it has a positive slope. This indicates that a positive relationship exists between the number of tomatoes on a plant and the hours of sunlight it receives.

Therefore, you can see that a scatter diagram is a graphical tool that helps us decide if two variables are related. Can we also have other types of relationships between two variables, or no relationship at all? Yes, absolutely.

A scatter diagram showing a negative relationship has a downward trend. In other words, the slope of the best-fit trend line is negative or pointing in the southeast direction, like in the image below.



No relation : The variable is said to be unrelated when the points are haphazardly scattered over the graph and do not show any specific pattern. .



Histogram

A histogram is a plot that lets you discover, and show, the underlying frequency distribution (shape) of a set of [continuous](#) data. This allows the inspection of the data for its underlying distribution (e.g., normal distribution), outliers, skewness, etc. An example of a histogram, and the raw data it was constructed from, is shown below:

How do you construct a histogram

1. On the vertical axis, place frequencies. Label this axis "Frequency".
2. On the horizontal axis, place the lower value of each interval. ...
3. **Draw** a bar extending from the lower value of each interval to the lower value of the next interval.

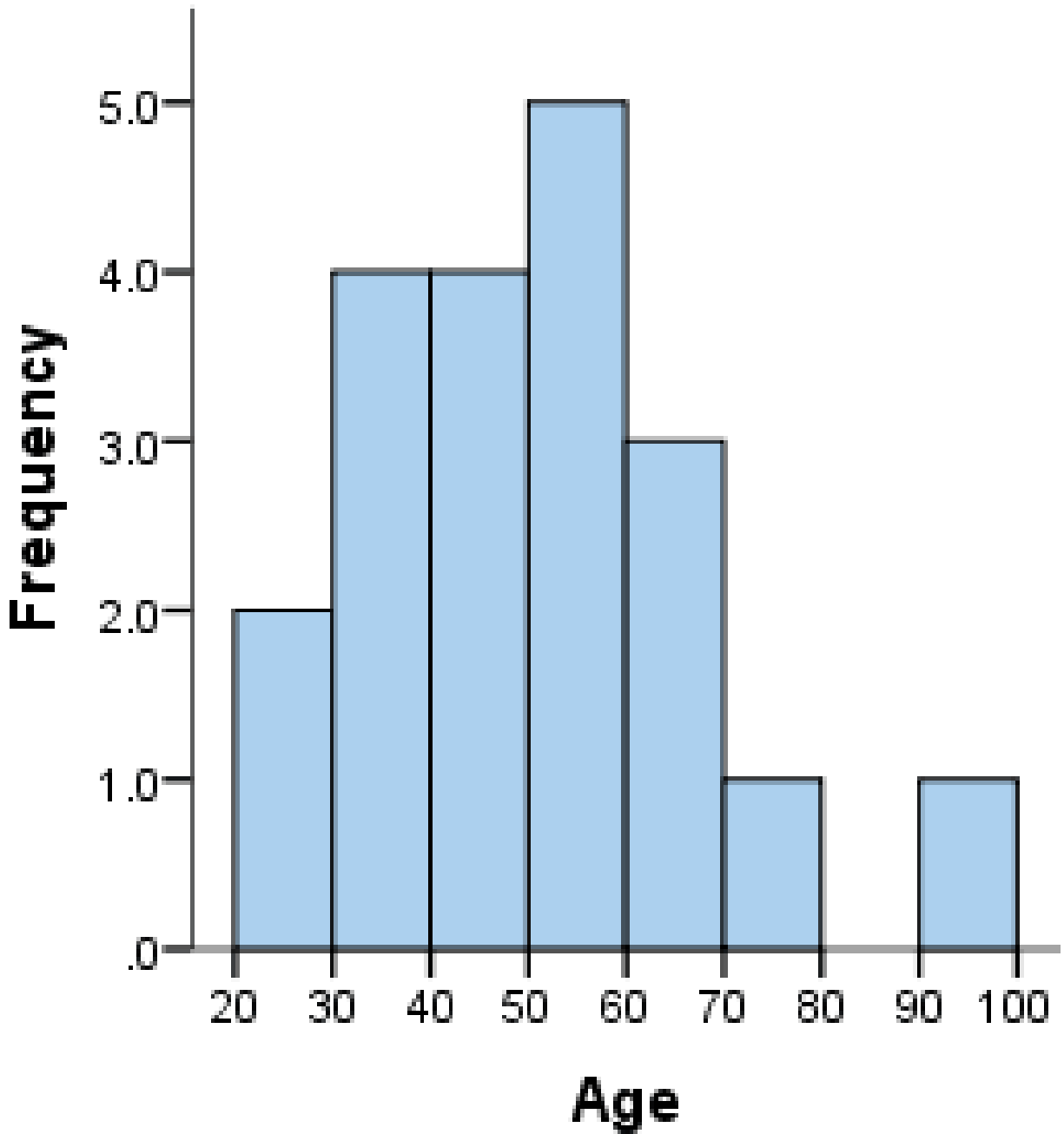
Problem

Bin	Frequency
-----	-----------

20-30	2
30-40	4
40-50	4
50-60	5
60-70	3
70-80	1
80-90	0
90-100	1

Draw a Histogram?

To construct a histogram from a continuous variable you first need to split the data into intervals, called **bins**. In the example above, **age** has been split into bins, with each bin representing a 10-year period starting at 20 years. Each bin contains the number of occurrences of scores in the data set that are contained within that bin. For the above data set, the frequencies in each bin have been tabulated along with the scores that contributed to the frequency in each bin (see below):



Notice that, unlike a bar chart, there are no "gaps" between the bars (although some bars might be "absent" reflecting no frequencies). This is because a histogram represents a continuous data set, and as such, there are no gaps in the data (although you will have to decide whether you round up or round down scores on the boundaries of bins).

What is the difference between a bar chart and a histogram?

The major difference is that a histogram is only used to plot the frequency of score occurrences in a continuous data set that has been divided into classes, called bins. Bar charts, on the other hand, can be used for a great deal of other types of variables including ordinal and nominal data sets.

Frequency Polygon

A frequency polygon is a graphical form of representation of data. It is used to depict the shape of the data and to depict trends. It is usually drawn with the help of a histogram but can be drawn without it as well. A histogram is a series of [rectangular](#) bars with no space between them and is used to represent [frequency distributions](#).

Problem#1: Weekly Income (in Tk) 100-300 300-500 500-700 700-900

Number of labours 30 40 20 25

(i) Draw a frequency polygon.

(ii) Find the class interval and class mark whose frequency is 25.

(iii) How many labourers have a weekly income of at least Tk 500 but not more than Tk 700?

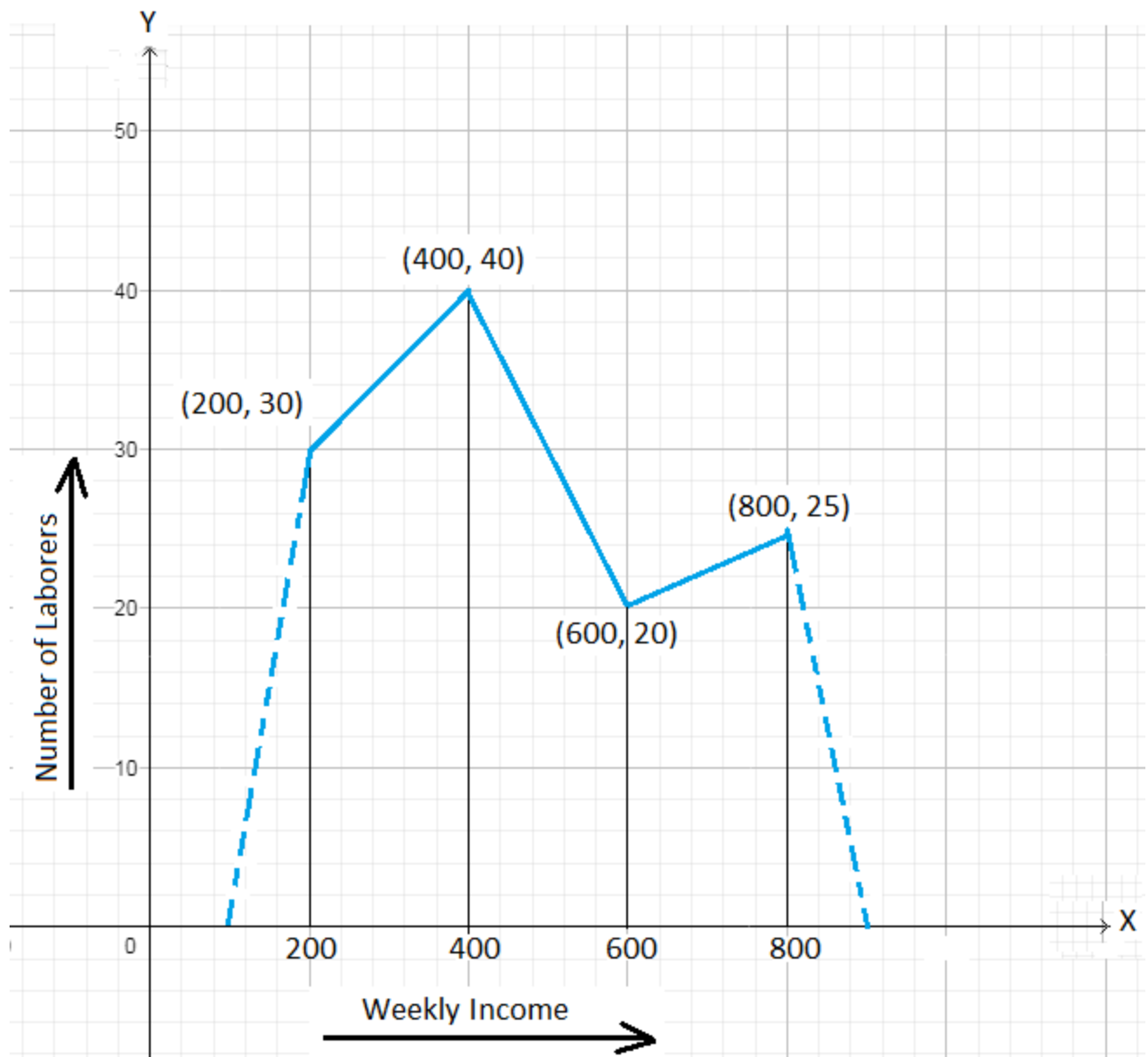
(iv) What is the range of weekly income of the largest number of labours?

Solution:

(i) Answer the following.

Weekly income (inTk)	Number of labours(fi)	Mid value(xi)
100-300	30	200
300-500	40	400
500-700	20	600
700-900	25	800

The following frequency polygon displays the weekly incomes of laborers of a factory.



[Save](#)

(ii) The frequency 25 corresponds to the class interval is 700-900 and class mark 800. The formula for class mark is:

$$\text{Class mark} = (\text{Upper limit} + \text{Lower limit}) / 2 = (900 + 700) / 2 = 800$$

(iii) The number of labours has to fall in the class interval 500 – 700 whose class mark is 600. The frequency corresponding to the class mark 600 is 20. Hence, the required number of labours is 20.

(iv) The largest number of labours belong to the class interval whose class mark is 400. The corresponding class interval is, (300 – 500). So, the largest numbers of labourers have a weekly income of at least Tk 300 but less than TK 500.

Problem#2: As the class marks of consecutive overlapping class intervals are 5, 15, 25, 35, 45, 55 we find the class intervals are 0 – 10, 10 – 20, 20 – 30, 30 – 40, 40 – 50, 50 – 60. Therefore, the frequency table is constructed as below.

Class Interval	Frequency
0 - 10	10
10 - 20	18
20 - 30	14
30 - 40	26
40 - 50	8
50 - 60	18

(i) Draw a histogram and also find frequency polygon .

(ii) What is the frequency of the class interval whose class mark is 15?

(iii) What is the class interval whose class mark is 45?

Steps to Draw a Histogram and Frequency Polygon

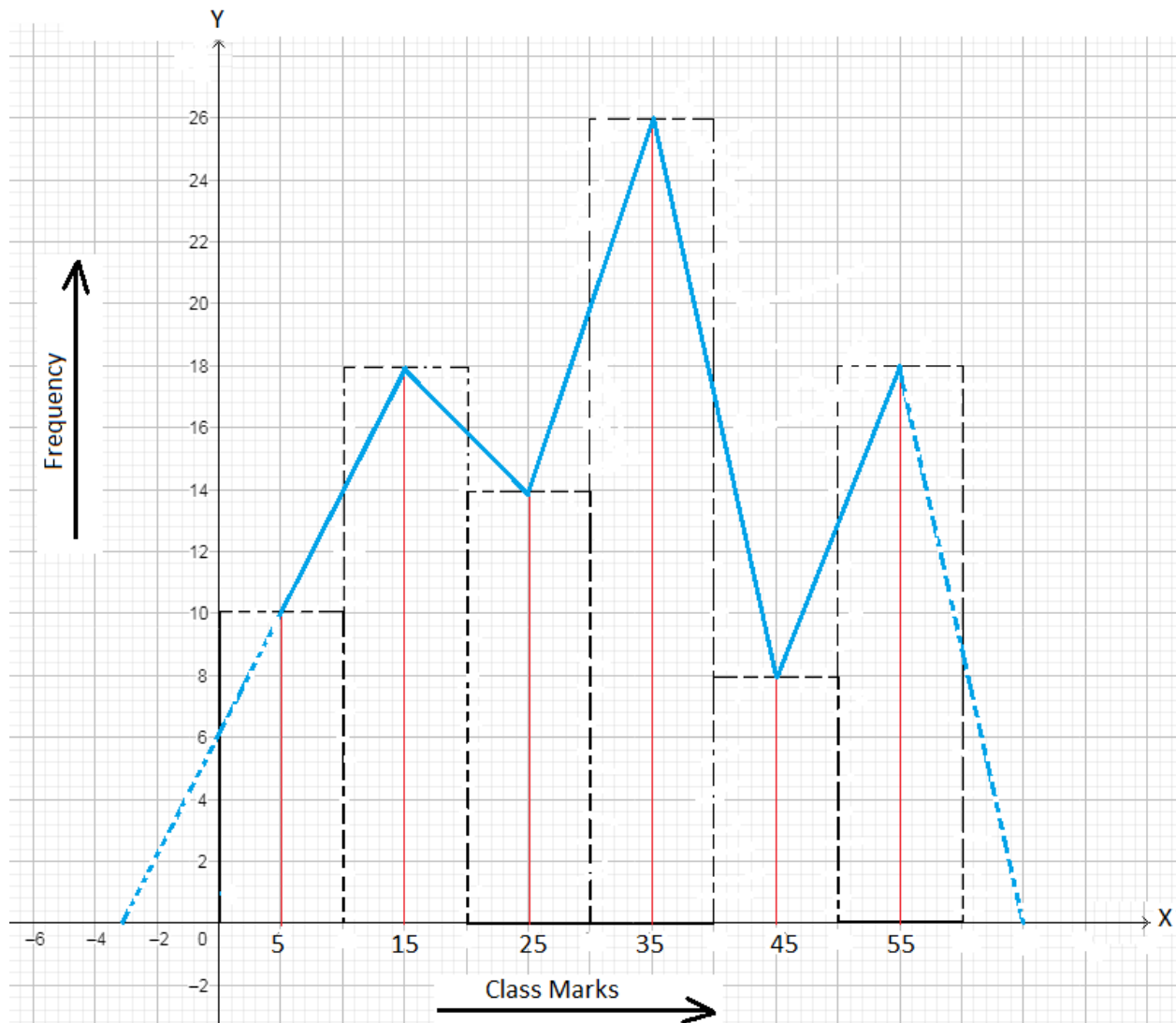
- Mark the class intervals for each class on the horizontal axis. We will plot the frequency on the vertical axis.
- Draw a Histogram.

- Calculate the class mark (mid value) for each class interval. The formula for class mark is:

$$\text{Class mark} = (\text{Upper limit} + \text{Lower limit}) / 2$$

- Mark all the class marks on the horizontal axis. It is also known as the mid-value of every class.
- Corresponding to each class mark, plot the frequency as given to you. The height always depicts the frequency. Make sure that the frequency is plotted against the class mark and not the upper or lower limit of any class.
- Join all the plotted points using a line segment. The curve obtained will be kinked.

The following frequency polygon displays the weekly incomes of laborers of a fact



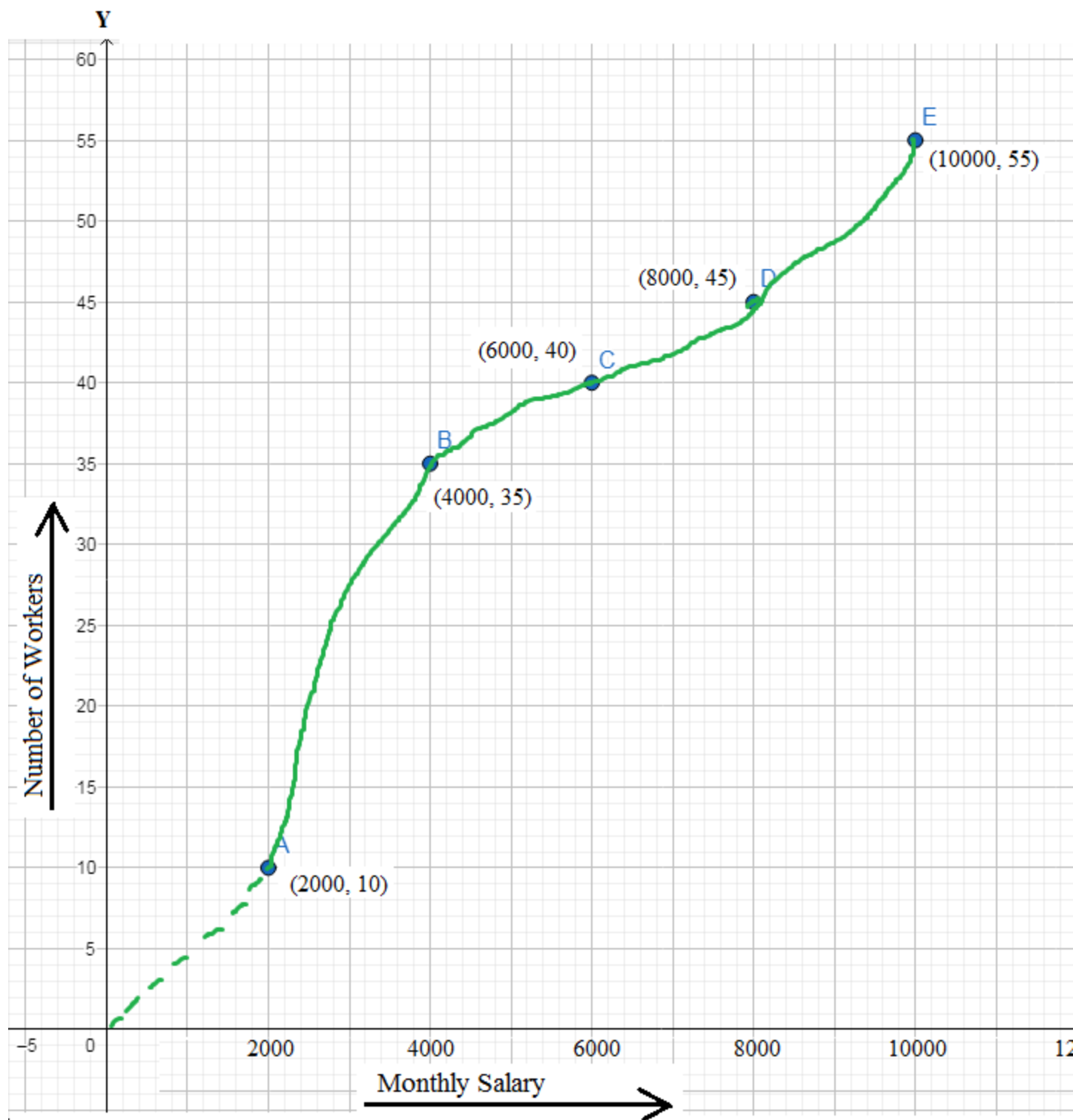
Cumulative frequency polygon (Ogive):

An ogive (oh-jive), sometimes called a **cumulative frequency polygon**, is a type of **frequency polygon** that shows **cumulative frequencies**. In other words, the **cumulative** percents are added on the graph from left to right. An ogive graph plots **cumulative frequency** on the y-axis and class boundaries along the x-axis. There are two types of cumulative frequency curves i)less than cumulative curves ii)more than cumulative frequency curves

Problem # Draw a cumulative frequency polygon .

Monthly salary	No. of workers	Cumulative frequency
0-2000	10	10
2000-4000	25	35
4000-6000	5	40
6000-8000	5	45
8000-10000	10	55

The monthly salaries of 55 workers of a factory are displayed in the following ogive.



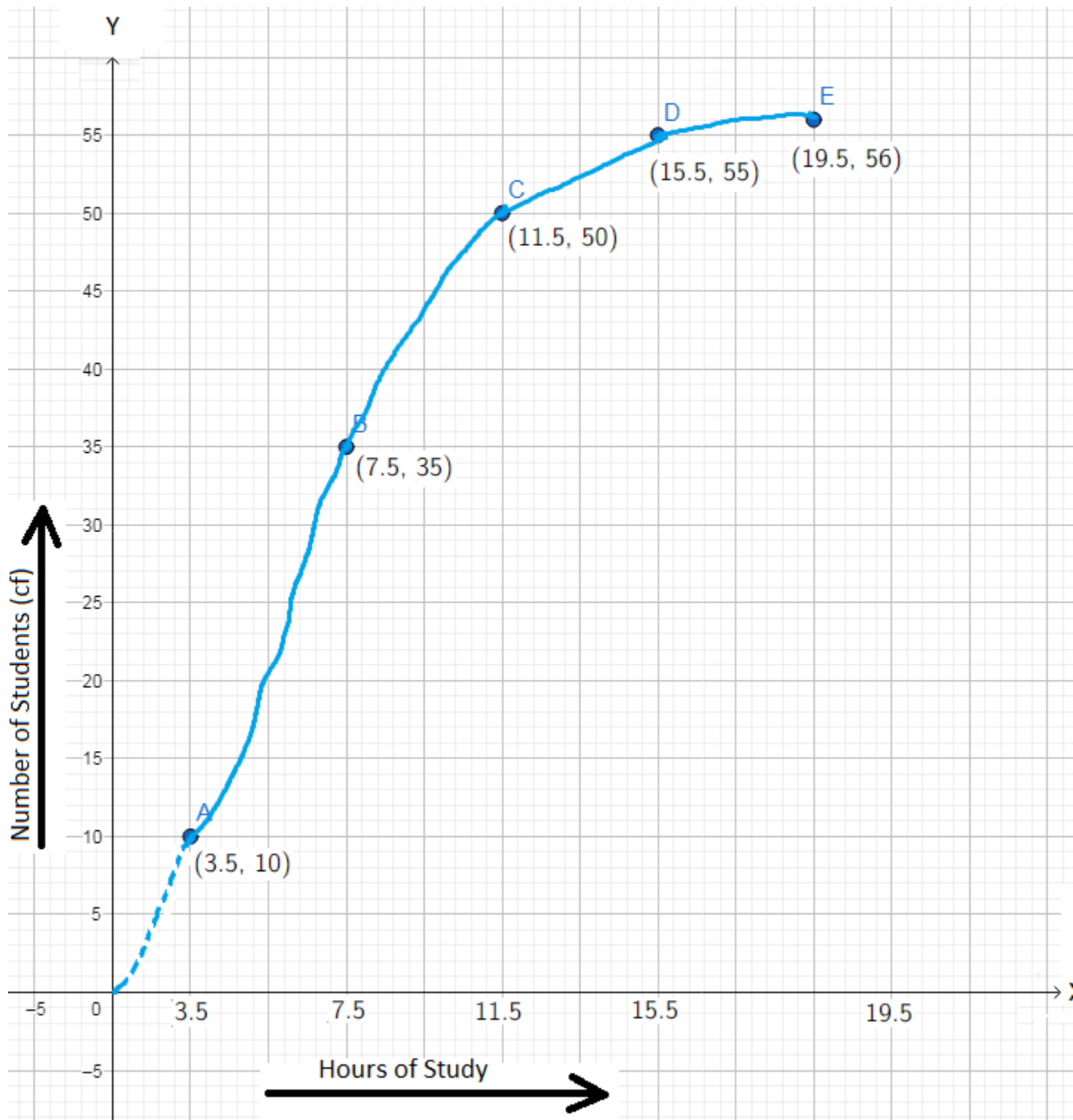
Problem #2 .Draw a cumulative frequency polygon .

Hour of study in a day	No.of students	Cumulative frequency
-0.5-3.5	10	10
3.5-7.5	25	35
7.5-11.5	15	50
11.5-15.5	5	55
15.5-19.5	1	56

Taking the upper limit of an interval and its corresponding cumulative frequency as x- and y-coordinates respectively, plot the points (3.5, 10)), (7.5, 35), (11.5, 50), (15.5, 55) and (19.5, 56).

Scale: On the x-axis, 1 cm = 4 hours.

On the y-axis, 1 mm = 1 student



Problem #3 .Construct the more than cumulative frequency table and draw the Ogive for the below-given data.

Marks	1-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80
Frequency	3	8	12	14	10	6	5	2

Solution:

Marks	1-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80
Frequency	3	8	12	14	10	6	5	2
Cumulative frequency (more than)	60	57	49	37	23	13	7	2

Plotting an Ogive:

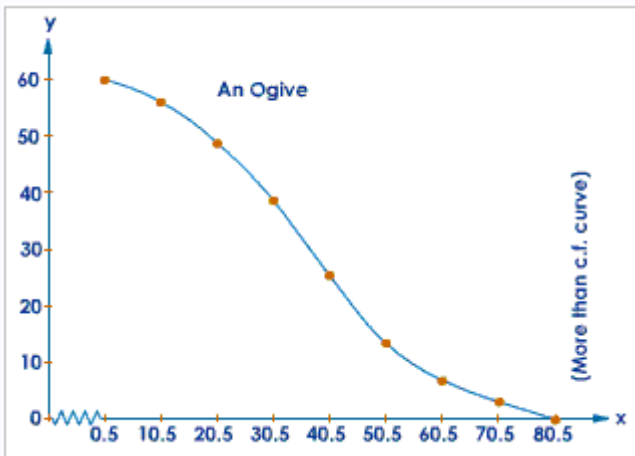
Plot the points with coordinates such as (70.5, 2), (60.5, 7), (50.5, 13), (40.5, 23), (30.5, 37), (20.5, 49), (10.5, 57), (0.5, 60).

An Ogive is connected to a point on the x-axis, that represents the actual upper limit of the last class, i.e., (80.5, 0)

Take x-axis, 1 cm = 10 marks

Y-axis = 1 cm – 10 c.f

Ogive Curve(more than cumulative frequency curve)



Relative frequency polygon: A **relative frequency polygon** has peaks that represent the percentage of total data points falling within the interval.

Relative frequency (R f): $\frac{\text{Class frequency}}{\text{Total frequency}} \times 100$

Problem#1: Weekly Income (in Tk) 100-300 300-500 500-700 700-900

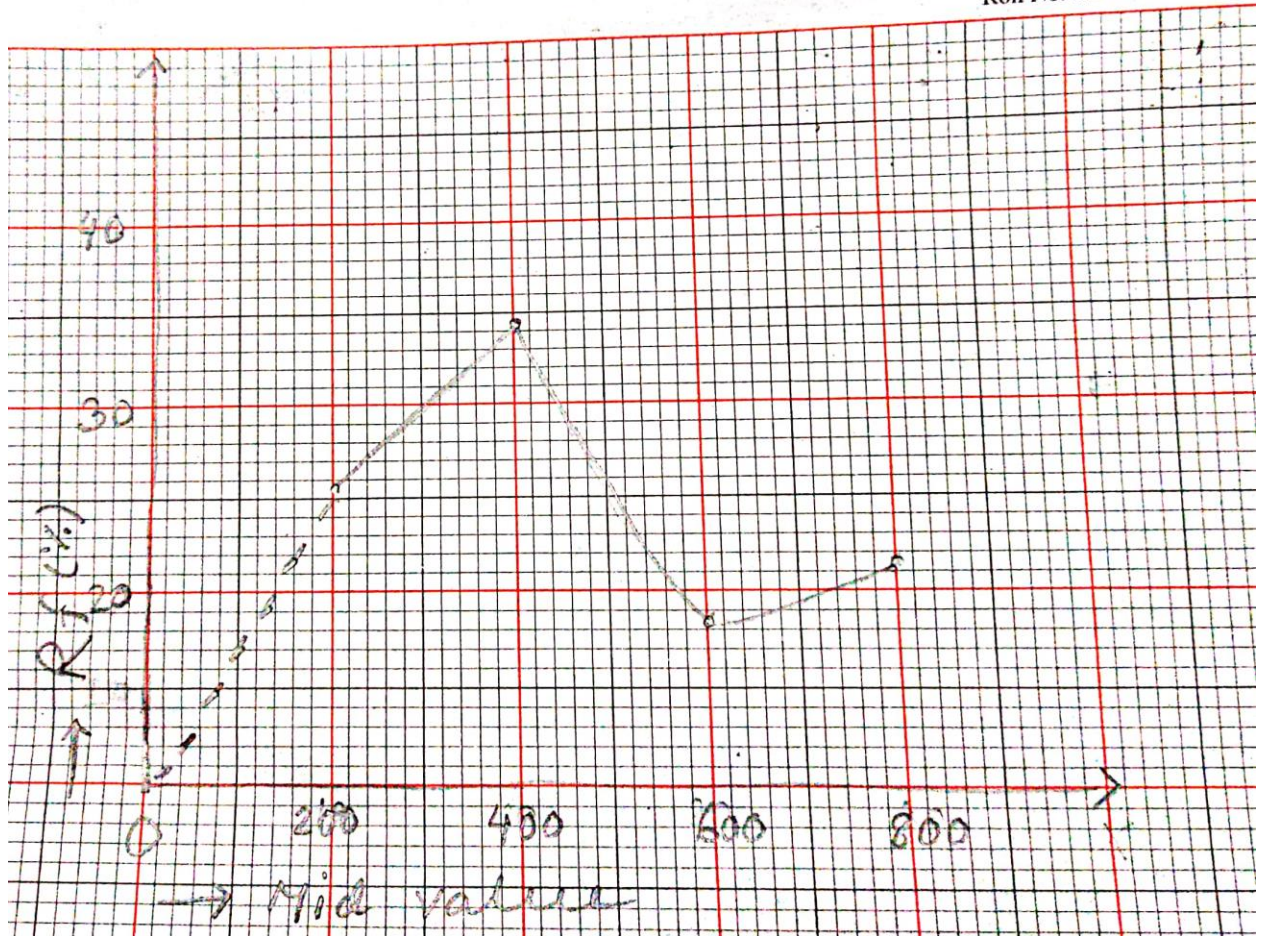
Number of labours 30 40 20 25

Draw a Relative frequency polygon.

Solution:

Answer the following.

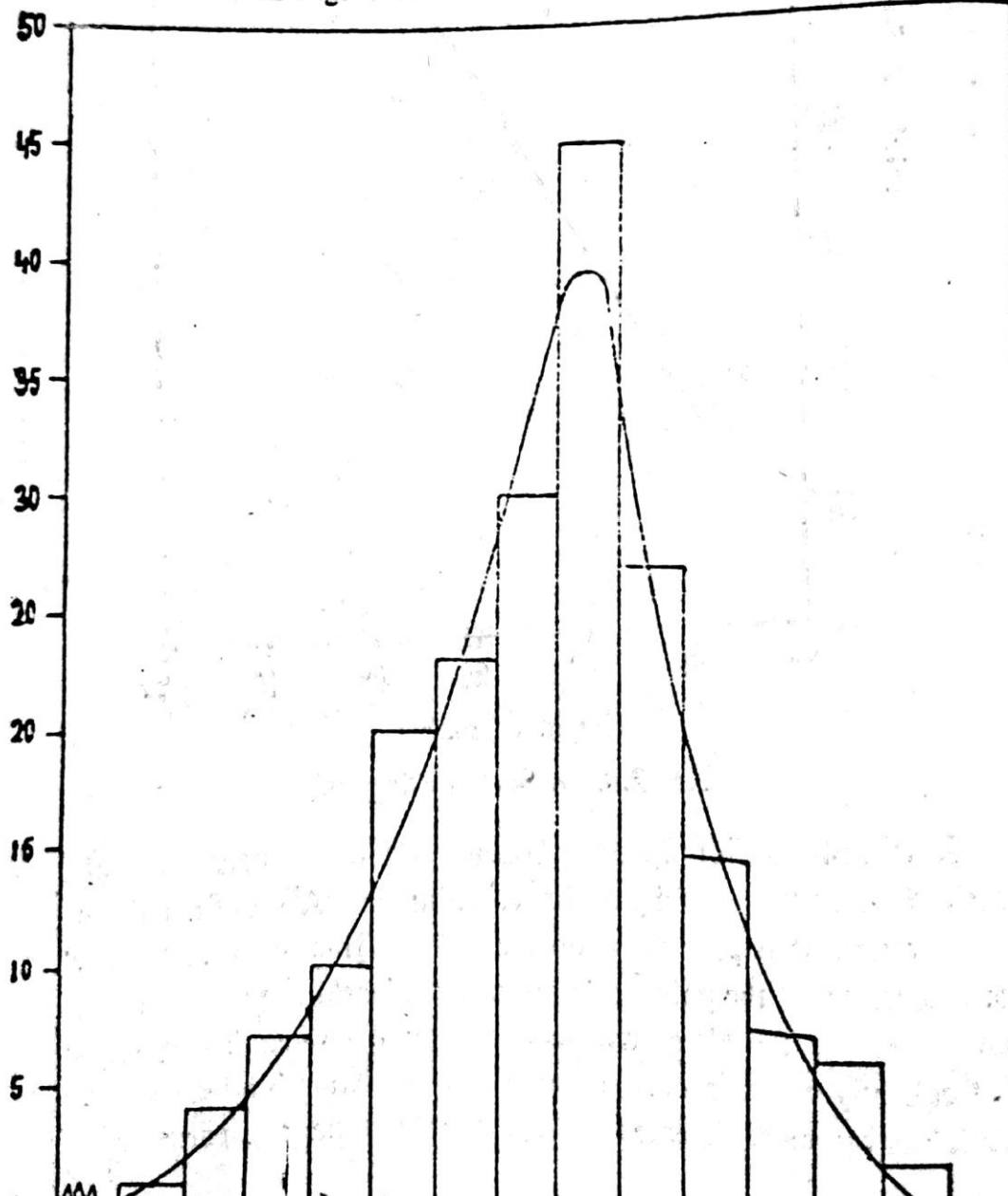
Weekly income (inTk)	Number of labours	Mid value	Relative frequency (%)
100-300	30	200	26.08 (30/115)x100
300-500	40	400	34.78
500-700	20	600	17.39
700-900	25	800	21.73
Total	115		



Relative frequency curve

2.6. FREQUENCY CURVES AND OGIVES

As mentioned earlier, the observations at our disposal usually constitute a sample from a population which is often infinite. It is reasonable to suppose that if the number of observations were very much larger and if the class intervals were made much narrower, the frequency polygon and the histogram would approach more and more closely to a smooth curve. This limiting curve is called the **frequency curve** of the population, and is of great theoretical importance. (The frequency curve for data in table 2.6 is shown in Fig. 2.9.) Similarly, if the number of observations is increased and at the same time the class intervals are made smaller, the cumulative frequency polygon will approximate to a smooth curve. This ideal limiting curve is called a **cumulative frequency curve** or an **ogive**. This is illustrated in Fig. 2.10. The S-shape is characteristic of most



PRESENTATION OF STATISTICAL DATA

ogives and arises whenever the frequency has a maximum well inside the range of the variable.

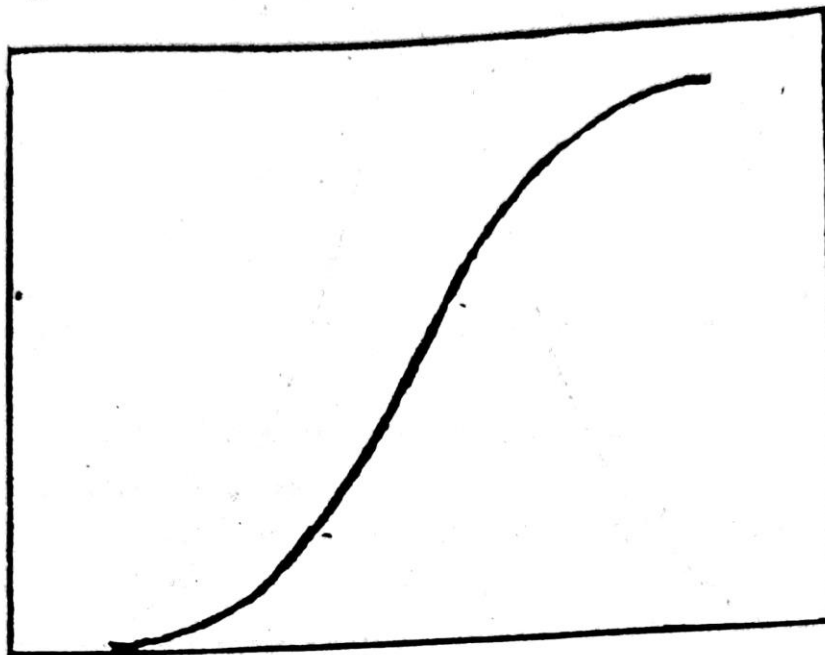


Fig. 2.10. An Ogive Curve.

Although a frequency curve or an ogive is often approximated by smoothing the corresponding frequency polygon or cumulative frequency polygon, it should be emphasised that the concept of a frequency curve or ogive is applicable **only** to a population rather than a sample. Moreover, strictly speaking, the concept of a continuous curve applies to a continuous variable only, but often it is used in connection with discrete variables as well.

Types of Frequency Curves.

There are many shapes of frequency curves, but some shapes are more common than others. We indicate below some of the common types of frequency curves that are met with in practice. These types are illustrated in Fig. 2.11.

(a) Bell-Shaped Frequency Curve.

A bell-shaped frequency curve is characterised by the fact that it has a maximum at the centre and the class frequencies decrease to zero symmetrically on either side of the maximum.

A frequency distribution is said to be **symmetrical** if its frequency curve can be folded along a vertical line so that the two halves of the figure coincide. A bell-shaped curve is necessarily symmetrical, but a symmetrical curve need not in general be

bell-shaped, although the two terms are sometimes used synonymously. A bell-shaped curve is highly important from the theoretical standpoint.

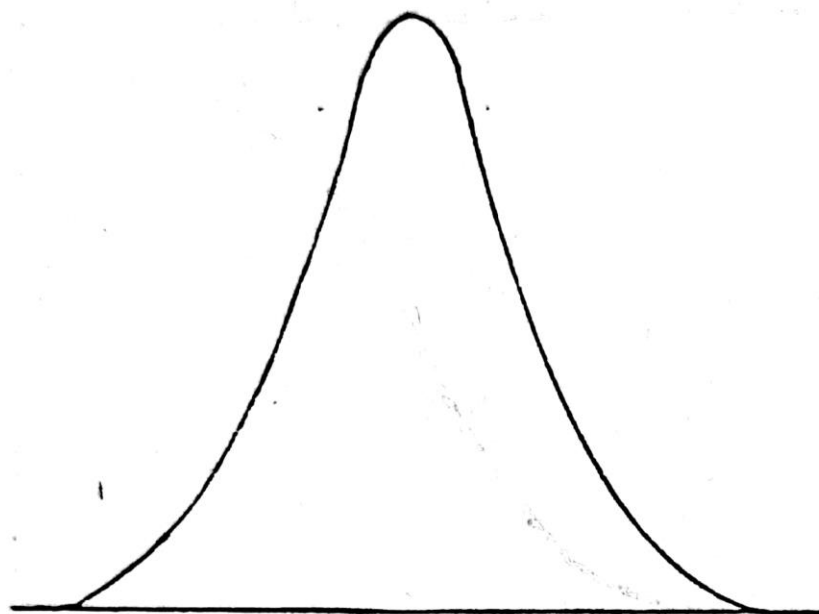


Fig. 2.11(a). A Bell-Shaped Frequency Curve.

(b) Moderately Asymmetrical (Skew) Frequency Curve.

A frequency curve is said to be asymmetrical or skew if it is lacking in symmetry. In a skew frequency curve, the tail of the curve to one side of the central maximum is longer than to the other. If the longer tail occurs to the right the curve is said to be **positively skew**, while if the reverse is true, the curve is said to be **negatively skew**. This (moderately skew) is the most common of all frequency curves.

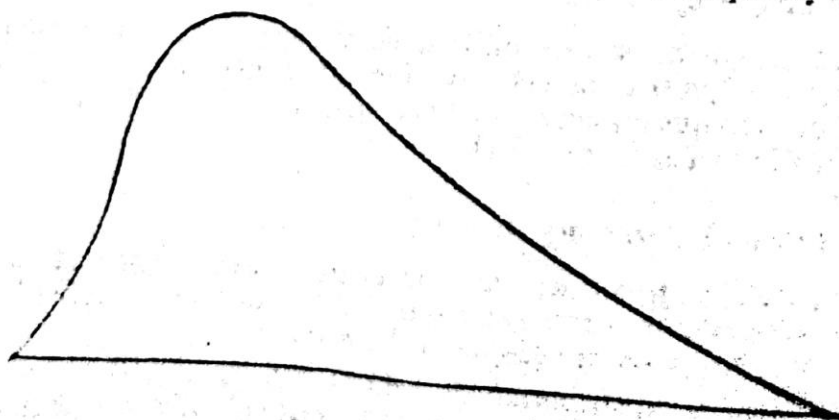


Fig. 2.11(b). Positively Skew Frequency Curve.

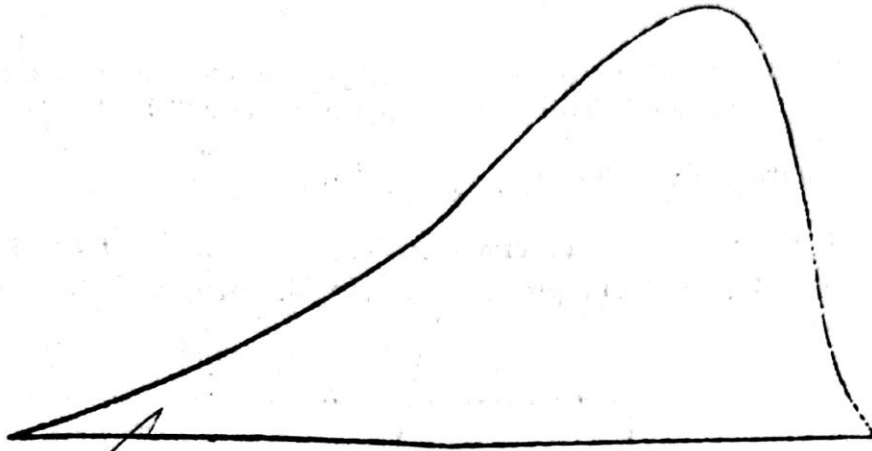


Fig. 2.11(c). Negatively Skew Frequency Curve.

(c) J-shaped Frequency Curve.

These frequency curves are highly skew. In J-shaped and reverse

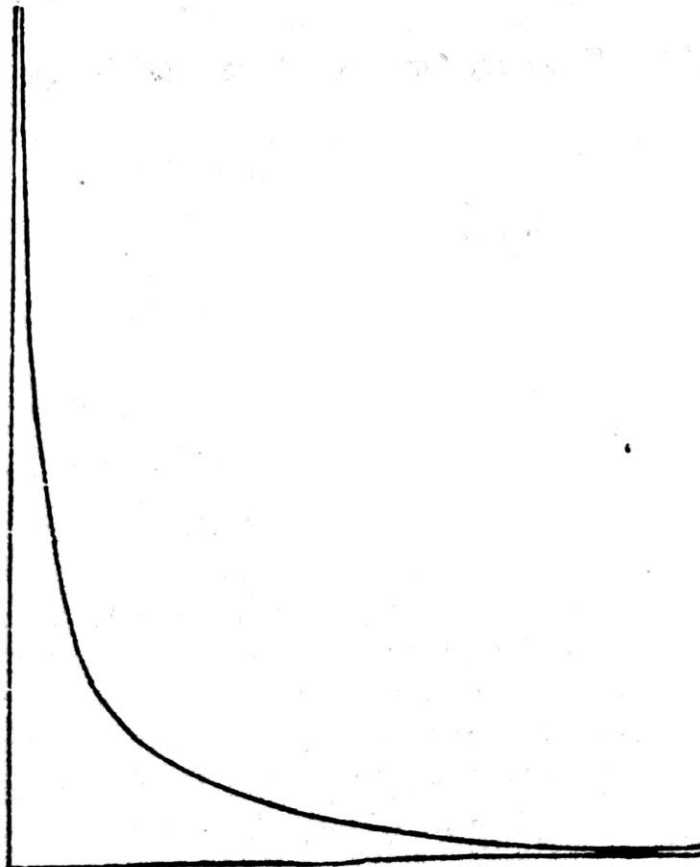


Fig. 2.11(d). J-Shaped Frequency Curve.

J-shaped curves, the maximum occurs at one end of the distribution. This form of distribution occurs in economic statistics.

(d) Frequency Curve for Rectangular Distributions.

In a rectangular or uniform distribution all the class frequencies are equal. Fig. 2.11(e) represents a rectangular distribution.

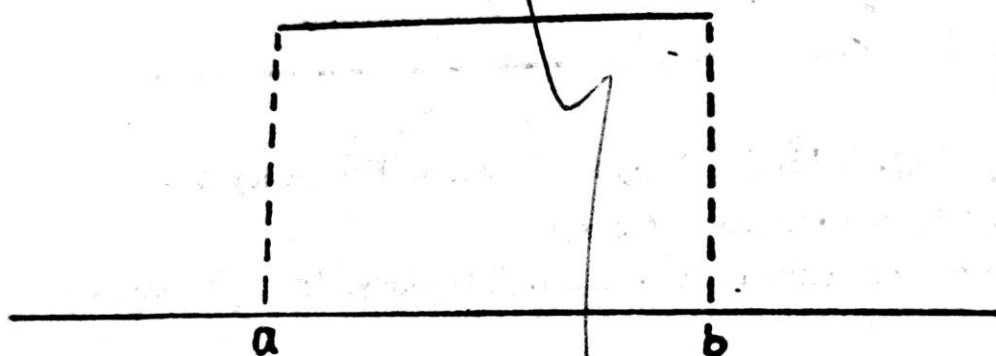


Fig. 2.11(e). Frequency Curve for Rectangular Distribution.

Questions and Problems

1. What is data?
2. What is variable? Discuss the types of variable.
3. Define primary data. State the various methods of collecting primary data.
4. Distinguish between primary data and secondary data.
5. What is frequency? What do you mean by a frequency distribution? How will you prepare a frequency distribution from raw data?
6. How to collect primary data by email and interview methods?
7. Discuss the different parts of a frequency table.
8. Show the differences between Histogram and Bar diagram.
9. What are the differences between histogram and frequency polygon?
10. State the various ways of presenting a frequency distribution graphically and discuss one of them.
11. Distinguish between discrete and continuous frequency distribution.
12. What is a pie chart?
13. What points should be considered in designing a good questionnaire?
14. Explain clearly the role and limitations of diagrams and graphs in presenting business data.
15. Write short notes:
 - (i) Stem-and-leaf Diagram;
 - (ii) Relative frequency polygon.
 - (iii) Secondary data;
 - (iv) Frequency curves

Problem:1. The daily wages (Taka) of 40 workers are given below:

88	204	108	68	101	129	119	149	210	103
96	104	146	127	97	89	123	108	94	149
93	118	136	187	102	191	148	163	172	93
104	132	131	142	92	87	144	183	105	195

- (i) Construct a continuous frequency distribution with suitable size of class interval.
- ii) Draw an ogive on the above frequency distribution

Problem#2.The marks of Management Trainee Assessment Exam are given below:

40	75	35	54	42	80	09	28
72	88	57	84	17	60	68	60
82	93	60	40	93	92	73	63
83	49	79	30	43	32	41	64
92	23	33	32	44	50	70	57

- (i) Taking first class as 0 – 9, construct a frequency distribution table from the above data.
- (ii) Construct cumulative frequency curve by “Less than” method.
- (iii) Draw a frequency polygon with the help of histogram.

Problem #3. You are given the following frequency distribution regarding annual profits in thousands taka of 115 firms:

Profit (000 Tk)	No. of firms
25 – 30	5
30 – 35	16
35 – 40	26
40 – 45	30
45 – 50	20
50 – 55	12
55 – 60	6

Construct less than and more than ogive curve and hence find the number of firms having profit between Tk 30000 and Tk 55000.

.....

.....